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FACULTY OF ELECTRICAL AND ELECTRONIC ENGINEERING

MACHINE LEARNING COURSE

Machine Learning Algorithm for Human Activity Recognition and Elderly Fall Detection Using MPU6050 Sensor Data

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MACHINE LEARNING COURSE REPORT

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Abstract

One of the main causes of health deterioration or mortality among the elderly is falls. The risk of falling increases proportionally with age and often occurs unexpectedly, making it difficult to prevent using technology-based systems. Nevertheless, early fall detection enables timely medical intervention during the golden hour for many types of injuries. This paper proposes a fall detection system for the elderly based on sensors and machine learning algorithms, which sends early alerts to caregivers. The system achieves an accuracy of 91.56% when using the Random Forest and 89.78% with Logistic Regression on a self-collected dataset. Overall, the Random Forest model achieved the highest accuracy among all tested methods. However, Logistic Regression proved to be more practical for real-world deployment due to its significantly smaller model size while maintaining competitive performance.

Keywords: Machine Learning, Healthcare System, Fall Detection.

Table of Contents

1. PROBLEM SITUATION AND PROBLEM DEFINITION 6
   1.1. *Problem situation* 6
   1.2. *Problem definition* 6
2. STATE OF ART 7
   2.1. *Exiting products/solution* 7
   2.2. *Scientific/engineering publication/whitepapers* 8
3. OBJECTIVES 9
   3.1. *General objectives* 9
   3.2. *Specific objectives* 9
4. PROPOSED SOLUTION 10
   4.1. *Description of proposed solution/design* 10
      4.1.1. Block diagram 10
      4.1.2. Operation description 10
      4.1.3. Proposed devices and components 10
   4.2. *Solution limitations* 10
   4.3. *Expected results* 11
   4.4. *Design approach and implementation methodology* 11
5. POTENTIAL IMPACT AND CONSIDERATION 13
   5.1. *Technical viability* 13
   5.2. *Economic feasibility* 13
   5.3. *Social viability* 13
   5.4. *Operational viability* 13
   5.5. *Environmental feasibility* 13
   5.6. *Alternatives solution/approach* 14
   5.7. *Codes of ethics (SAE)* 14
6. WEEKLY PROJECT DEVELOPMENT SCHEDULE 15
   6.1. *Schedule and plan* 15
   6.2. *Member roles* 16
7. PROJECT PROGRESS 17
   7.1. *Human activity data collection device using MPU6050 sensor* 17
      7.1.1. Theoretical basis 17
      7.1.2. The MPU6050 implement analysis 17
      7.1.3. Results 18
   7.2. *Human activity recognition dataset (HARDataset)* 19
      7.2.1. Theoretical basis 19
      7.2.2. Features extraction analysis 19
      7.2.3. Results 21
   7.3. *Machine Learning models* 25

7.3.1. Theoretical basis 25
7.3.2. Model analysis 25
7.4. Evaluation metrics 25
8. RESULTS AND DISCUSSION 27
8.1. Results 27
8.2. Evaluation 28
8.3. Real time Implementation Results 31
8.4. Discussion 31
9. RISK ANALYSIS 33
9.1. Risk factor inputs 33
9.1.1. Risk factors 33
9.1.2. Probability and impact scales 33
9.2. Risk analysis result 33
9.2.1. Levels of risks 33
9.2.2. Priority risks 35
10. FINANCIAL AND ECONOMIC REPORT 36
11. STANDARDS 37
12. CONCLUSION 39
13. REFERENCES 41
14. APPENDICES 43

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List of Figures

Figure 1: Apple Watch Series 4. 7
Figure 2: iVi Intelligent Pendant 8
Figure 3: Block diagram 10
Figure 4: Flow diagram 12
Figure 5: MPU6050 output values 17
Figure 6: Data collection device's webserver interface. 18
Figure 7: Data samples for Standing and Fall activities 20
Figure 8: Distribution of Labels in Training and Testing dataset. 20
Figure 9: Matrix of Correlations among Features 21
Figure 10: Features importance for models 22
Figure 11: Histogram of Features 23
Figure 12: Scatter plot of PCA and t-SNE features on features data.. 24
Figure 13: Average KNN performance across 5 folds due to K value. 27
Figure 14: Average Random Forest performance across 5 folds due to n_estimators 28
Figure 15: Confusion matrix of models on test dataset 30
Figure 16: Real time implementation 31
Figure 17: Risk matrix 34

List of Tables

Table 1: Weekly schedule to complete project's objectives 15
Table 2: Member roles 16
Table 3: Models performance on test dataset 29

CHAPTER 1: PROBLEM SITUATION AND PROBLEM DEFINITION

1.1.Problem situation

Falls are among the leading causes of injuries and mortality in the elderly population. Each year, there are about 3 million emergency department visits and about 1 million fall-related hospitalizations among older adults [1]. Falls are the cause of hip fractures at nearly 319,000 older people [2], and are the most common cause of traumatic brain injuries (TBI) [3]. With the aging of society, the need for continuous monitoring of daily activities to ensure safety and timely medical support has become increasingly important. Traditional manual supervision is labor-intensive and impractical for round-the-clock monitoring, while unexpected falls occur without warning, making prevention and timely response difficult. Therefore, there is a growing demand for intelligent monitoring systems capable of automatically recognizing human actions and detecting falls in real-time.

1.2.Problem definition

The problem is to design and implement an automatic monitoring system that can accurately identify human activities and detect fall events, especially among elderly individuals. The system should leverage sensor data and machine learning algorithms to distinguish normal activities from abnormal ones such as falls. Moreover, the solution must achieve high accuracy, minimize false alarms, and provide timely alerts to caregivers or medical staff, ensuring rapid intervention during the critical golden hour of treatment.

CHAPTER 2: STATE OF ART

2.1. Exiting products/solution

The Apple Watch Series 4 is an important example of how wearable devices use sensors and machine learning for activity recognition and fall detection. It is equipped with a high-sensitivity accelerometer (up to 32 g) and a gyroscope, forming an IMU similar to the MPU6050 but more advanced. With these sensors, the watch can recognize many activities such as walking, running, cycling, swimming, or climbing stairs, and it can also automatically detect when a workout starts. The highlight of the device is its Fall Detection function. By

Figure 1: Apple Watch Series 4 [4].

analyzing sudden acceleration and changes in body orientation, the watch compares motion patterns with trained fall data to decide if a fall has occurred. When this happens, the watch vibrates, sounds an alarm, and shows an alert. If the user does not respond and stays immobile for about one minute, it automatically calls emergency services and shares the location with emergency contacts. Fall events are also stored in the Health app for later review. This combination of accurate sensing, action recognition, and automatic emergency response shows how consumer wearables can support both daily activity tracking and important health protection.

The iVi Intelligent Pendant is a lightweight wearable fall detector designed for elderly care. It uses an accelerometer to identify sudden movements consistent with a fall. When a potential fall is detected, the device starts a pre-alarm phase (green LED and audio alert), giving the wearer ~10 seconds to cancel if no help is needed. If not canceled, it escalates to a red alarm and transmits a signal to a connected telecare base unit, which notifies monitoring centers for assistance. Unlike smartwatches (e.g., Apple Watch), the iVi cannot directly call emergency services but relies on a

linked home telecare unit. Its design focuses on reliability, long battery life, and simplicity, making it widely adopted in elderly monitoring solutions.

Figure 2: iVi Intelligent Pendant [5].

2.2. Scientific/engineering publication/whitepapers

The article "SisFall: A Fall and Movement Dataset" by Sucerquia A (2017) [6] introduced SisFall, one of the largest publicly available datasets for fall detection and activity recognition. It contains accelerometer and gyroscope data collected from wearable sensors on elderly and young volunteers. The dataset has become a standard benchmark for testing fall detection algorithms.

The article "Fall Detection and Prediction Based on IMU and EMG Sensors for Elders" by W Siwadamrongpong (2022) [7] addresses fall prediction and detection in the elderly using Inertial Measurement Unit (IMU) sensors and Electromyogram (EMG) signals. Features from EMG were combined with IMU data and processed with machine learning models to classify gait patterns and fall events. Results showed that EMG-based models achieved moderate accuracy (~72% for 3 classes and ~68% for 4 classes), while IMU-based models using Random Forest performed significantly better, reaching 94.72% (3-class) and 87.70% (4-class) classification accuracy. These findings highlight the potential of IMU sensors for reliable fall detection.

CHAPTER 3: OBJECTIVES

3.1.General objectives

To develop an intelligent monitoring system that classifies human activities and detects falls in elderly individuals using sensor data and machine learning algorithms, thereby enhancing healthcare support and ensuring timely intervention.

3.2.Specific objectives

- To develop a data acquisition system using the MPU6050 sensor that ensures consistency and reliability across different subjects during activity recording.
- To design and implement machine learning models for classifying human activities and detecting falls with high accuracy.
- To evaluate and compare the performance of different algorithms (e.g., Random Forest, SVM, Deep Learning models) on the dataset.
- To integrate the fall detection module with an alert mechanism that notifies caregivers or healthcare providers in real time.
- To optimize machine learning algorithms for deployment on embedded devices, ensuring efficiency, low latency, and energy-aware performance.
- To design a scalable system architecture that can be extended for continuous monitoring in real-life healthcare environments.

CHAPTER 4: PROPOSED SOLUTION

4.1.Description of proposed solution/design

4.1.1. Block diagram

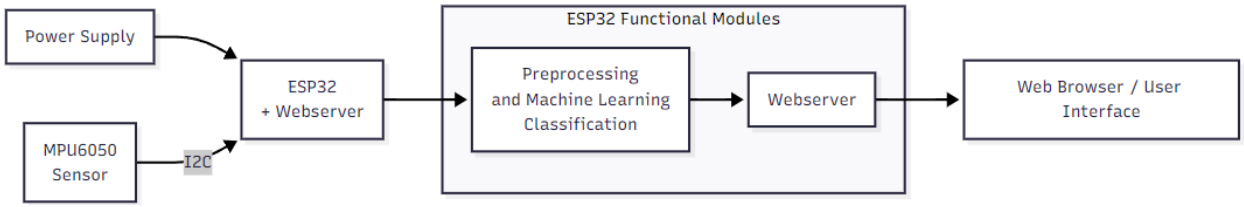

Figure 3: Block diagram.

4.1.2. Operation description

The data collected from the MPU6050 sensor are transmitted to the ESP32 via the I²C protocol. On the ESP32, a preprocessing algorithm is applied to the raw sensor signals before they are used as input to a machine learning–based classification model. The output of the classification results is then transmitted and visualized through the webserver.

4.1.3. Proposed devices and components

The proposed system is developed based on the following key hardware components:

- Sensor (MPU6050): A six-axis motion tracking device that integrates a 3-axis gyroscope and a 3-axis accelerometer [8]. It is used to capture motion and orientation data of the subject in real time.
- Microcontroller (ESP32): A low-power, high-performance microcontroller with integrated Wi-Fi and Bluetooth. It is responsible for acquiring sensor data through the I²C protocol, performing preprocessing, running the machine learning inference, and hosting the embedded webserver [9].
- Power Supply (Battery): A rechargeable battery unit that provides stable power to the ESP32 and the sensor module, ensuring system portability and continuous operation.

4.2.Solution limitations

Although the proposed system is cost-effective due to the use of low-priced components such as the MPU6050 sensor and the ESP32 microcontroller, several limitations remain:

- While inexpensive and widely available, the MPU6050 suffers from certain measurement inaccuracies and drift over time, which may affect the reliability of motion tracking.
- The ESP32 provides built-in wireless connectivity and sufficient performance for lightweight tasks. However, its limited RAM and CPU resources restrict the deployment of computationally intensive machine learning algorithms, making it more suitable for simple or optimized models.

4.3. Expected results

The proposed system is expected to operate in real time, enabling continuous acquisition and processing of motion data from the MPU6050 sensor. The embedded machine learning model deployed on the ESP32 microcontroller is anticipated to achieve an accuracy higher than 85% in classifying activities, while maintaining low latency suitable for real-time applications. Furthermore, the system has the potential to be further developed for elderly care applications, such as fall detection and activity monitoring, thereby contributing to improved safety and healthcare support.

4.4. Design approach and implementation methodology

The proposed system is designed following a hardware-software integrated approach. The MPU6050 sensor acquires both accelerometer and gyroscope data, which are transmitted to the ESP32 microcontroller via the I²C protocol. On the ESP32, a preprocessing module extracts statistical features from the raw sensor data, including mean, median, maximum, minimum, skewness, kurtosis, and variance, to serve as inputs for the machine learning model.

The machine learning model, either a Decision Tree or a Random Forest, is first trained offline on a computer with labeled datasets and then embedded on the ESP32 by transferring the trained parameters. The ESP32 also hosts an embedded webserver that visualizes the current user activity and provides real-time plots of sensor data and classification results.

System validation includes evaluating the model's accuracy, measuring latency to ensure real-time performance, and testing the stability of the system during continuous operation. This design methodology ensures that the system is capable of providing reliable real-time activity classification while remaining suitable for low-power, embedded deployment.

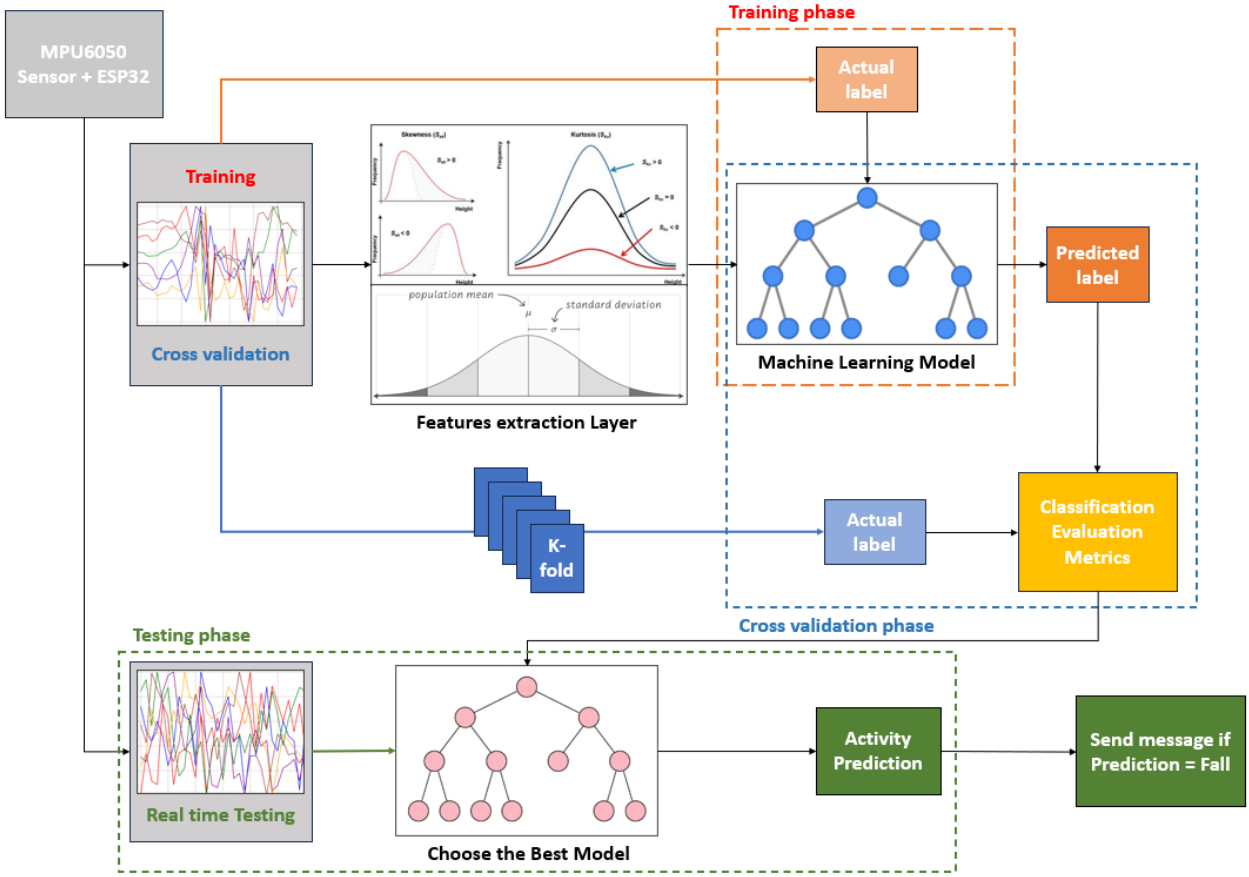

Figure 4: Flow diagram.

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CHAPTER 5: POTENTIAL IMPACT AND CONSIDERATION

5.1. Technical viability

The system is technically feasible since all the main components are well established: the MPU6050 sensor can accurately measure acceleration and angular velocity; the ESP32 has sufficient computational power to perform inference using lightweight ML models such as Decision Tree or Random Forest; and feature extraction methods (mean, median, max) help reduce computational load while ensuring real-time operation. In addition, the ESP32's wireless communication enables data transmission and real-time visualization of the results.

5.2. Economic feasibility

The system is cost-effective due to the use of widely available and inexpensive components: ESP32 (~5–7 USD), MPU6050 (~2–3 USD), rechargeable battery and wristband (~5–10 USD). The total cost of a prototype device is only around 15–20 USD, which is significantly cheaper compared to commercial smartwatches (100–300 USD). Therefore, the project is economically feasible for both research and trial deployment.

5.3. Social viability

The product targets elderly people and patients who need health monitoring, providing high social value by enabling timely fall detection and reducing risks. It can also be extended to personal healthcare and fitness applications. With its low cost and user-friendly design, the system is accessible to a wide range of users.

5.4. Operational viability

The system can operate stably for long periods with rechargeable batteries, lightweight software, and optimized ML models. Users only need to wear the device without complex interaction. Real-time performance with low latency (<200 ms) allows activity recognition and fall alerts to be delivered almost instantly. Thus, the project is operationally viable in practice.

5.5. Environmental feasibility

The device has low power consumption (ESP32 draws ~160 mA with Wi-Fi active, much less in sleep mode). The rechargeable battery reduces electronic waste. Moreover, using durable and reusable components extends the product's lifecycle, minimizing environmental impact.

5.6. Alternatives solution/approach

- Hardware alternatives: use more advanced IMUs (e.g., BNO055, MPU9250) for higher accuracy.
- Algorithmic alternatives: explore other lightweight models such as k-NN, SVM, or optimized deep learning models (TinyML).
- Deployment alternatives: instead of on-device inference, raw sensor data could be sent to a smartphone or server for processing, with only the results returned to the device.

5.7. Codes of ethics (IEEE)

The project follows IEEE's ethical guidelines:

- Responsibility: ensure the system provides accurate results and avoids false alarms that may endanger users.
- Privacy: protect personal data (sensor readings, activity patterns) and only share with user consent.
- Transparency: make the algorithm and alert mechanism understandable and clear to end-users.
- Sustainability: design for low power consumption and reduced electronic waste.
- Fairness: offer the solution at a low cost to make it accessible for a broad range of social groups.

CHAPTER 6: WEEKLY PROJECT DEVELOPMENT SCHEDULE

6.1.Schedule and plan

Table 1: Weekly schedule to complete project's objectives

Schedule / Objectives	W1 01-08/9	W2 08-15/9	W3 15-22/9	W4 22-29/9	W5-W8 29/9-27/10
Design ideas for the system, the data collection methods and data types, as well as the system's inputs and outputs.	X				
Investigate the previously proposed approaches and the current state of the art.	X				
Design and implement a human activity data collection device using MPU6050 sensor controlled via webserver interface.		X	X		
Deploy data collection on subjects.		X	X		
Explore methods for processing and feature extraction from time-series data.	X				
Implement a data processing pipeline to construct the dataset and extract features.			X	X	
Perform data visualization, analysis and feature selection.				X	
Implement machine learning algorithms from scratch and hyperparameters tuning.				X	
Embedded deployment on microcontroller and real-time system execution					X

6.2.Member roles

Table 2: Member roles.

Nguyễn Thanh Tùng - Leader	Nguyễn Bảo Long - Member
• Design a human activity data collection system using the MPU6050 sensor and an ESP32 microcontroller, control via a web server interface. • Deploy data collection on subjects • Data exploration, gain insights into the dataset via visualization and evaluate model performance. • Real time implementation for the system	• Write system code to organize the data, perform preprocessing, extract features, and implement machine learning classification models. • Deploy data collection on subjects • Feature selection and parameter selection to improve model efficiency, enhance accuracy through optimal data and hyperparameter choices. • Write the report

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CHAPTER 7: PROJECT PROGRESS

7.1. Human activity data collection device using MPU6050 sensor

7.1.1. Theoretical basis

The MPU6050 [8] has both 3-Axis accelerometer and 3-Axis gyroscope integrated on a single chip. The gyroscope measures rotational velocity or rate of change of the angular position over time, along the X, Y and Z axis, by using MEMS technology that exploit the Coriolis Effect [10]. On the other hand, the accelerometer measures gravitational acceleration along the 3 axes. It operates by measuring variations in capacitance created by tiny spring-supported masses that shift when moving [11]. By combining data from both sensors, the MPU6050 can track linear and rotational movements, making it highly useful in devices such as drones, mobile phones, robotics platforms, and motion analysis tools.

The MPU6050 sensor is also called six-axis motion tracking device, because of its 6 outputs: 3 accelerometer outputs and the 3 gyroscope outputs.

EXT_SYNC_SET	FSYNC Bit Location
0	Input disabled
1	TEMP_OUT_L[0]
2	GYRO_XOUT_L[0]
3	GYRO_YOUT_L[0]
4	GYRO_ZOUT_L[0]
5	ACCEL_XOUT_L[0]
6	ACCEL_YOUT_L[0]
7	ACCEL_ZOUT_L[0]

Figure 5: MPU6050 output values [6].

7.1.2. The MPU6050 implement analysis

The choice of the MPU6050 sensor for the human activity data collection device is based on several practical and technical considerations:

- Low power consumption [6], small packaged and lightweight, suitable for wearable or mobile health monitoring systems.
- The outputs offer comprehensive information about body movement, posture, and orientation, which are critical for reliable human activity recognition.

- The flexibility of outputs range ensures that the collected signals are easy to scale, filter, and map into meaningful activity labels.
- Cost-effective and available for mass production.

7.1.3. Results

The system was successfully implemented by integrating the MPU6050 sensor with the ESP32 microcontroller. The sensor collects acceleration and angular velocity data along three axes, which are then transmitted to the ESP32 via WiFi in Access Point (SoftAP) mode for processing. On the ESP32, a HTTP webserver was developed, allowing users to control the device and monitor the data collection in real time through a standard web browser.

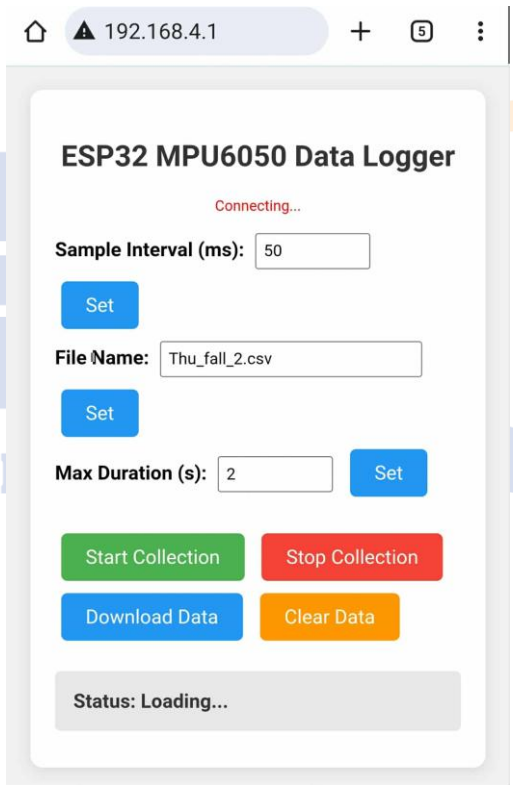

Figure 6: Data collection device's webserver interface.

Experimental results show that the MPU6050 sensor data was collected stably and displayed fully on the web interface. Additionally, the system demonstrated low power consumption, making it suitable for applications in human activity data collection devices.

7.2. Human activity recognition dataset (HARDataset)

7.2.1. Theoretical basis

The dataset utilized in this study is currently being collected by installing the device on the arm of 9 participants. The recorded dataset encompasses a set of predefined activity classes, namely: standing, lying, walking, stand-up, and falling. For each activity, a specific number of trials are conducted to ensure diversity and consistency in the captured signals. Each trial, referred to as a sample, is recorded in the form of time-series data, which is subsequently stored in an individual CSV file for ease of processing and organization.

To maintain clarity and systematic structuring, a standardized naming convention is adopted for all CSV files. Each file is named using the format Subject_Activity_Number, where:

- Subject: denotes the participant name.
- Activity: specifies the performed activity label.
- Number: represents the trial index corresponding to that activity for the given subject.

This convention not only facilitates straightforward identification of the data but also enables efficient indexing, retrieval, and analysis throughout the experimental and modeling stages.

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7.2.2. Dataset analysis

The human activity dataset which can be accessed on Hugging Face by searching for the repository HAR-Dataset [13], was collected from 10 subjects and divided into a training set (7 subjects) and a test set (3 subjects). Each subject performed common actions that typically occur before a fall in elderly individuals, such as walking (label 0), standing (label 1), and standup (label 3), as well as other actions like lying (label 2) and fall (label 4). Each action is represented by 6 continuous float values obtained from the accelerometer and gyroscope sensors of the MPU6050 module. The data samples representing standing and falling activities can be visuallize as shown in **Figure 7** (The raw data has been z-score scaled to better visualization).

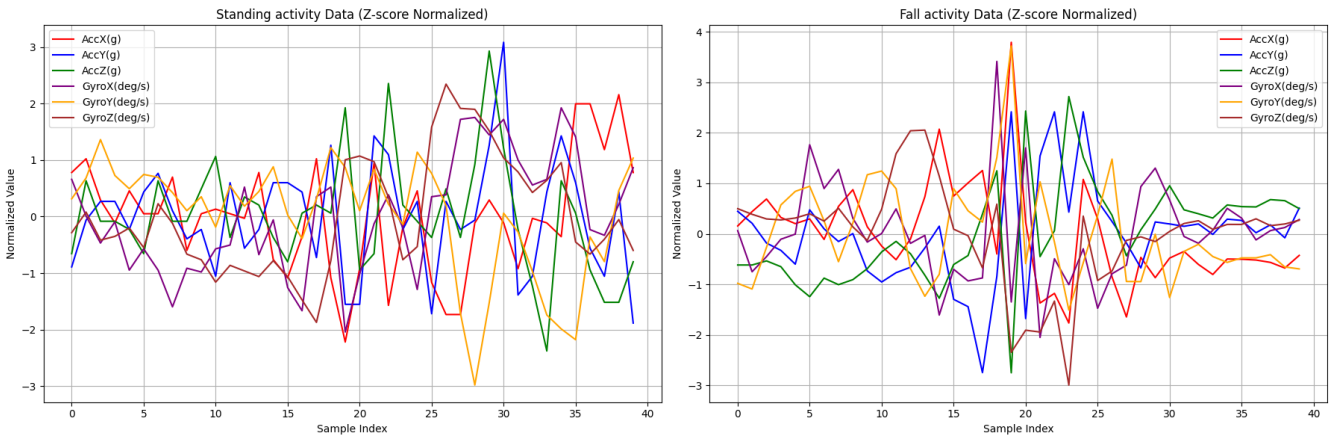

Figure 7: Data samples for Standing and Fall activities.

The **Figure 8** illustrates the distribution of activity labels in the training and test datasets. Each activity has an approximately equal number of samples in both the training and test sets, indicating a balanced dataset. The dataset maintaining a similar proportion of samples across all activity classes helps ensure that the model learns each action fairly, reducing potential bias toward any dominant class. Overall, the dataset distribution appears well-prepared, supporting stable and reliable model training and evaluation.

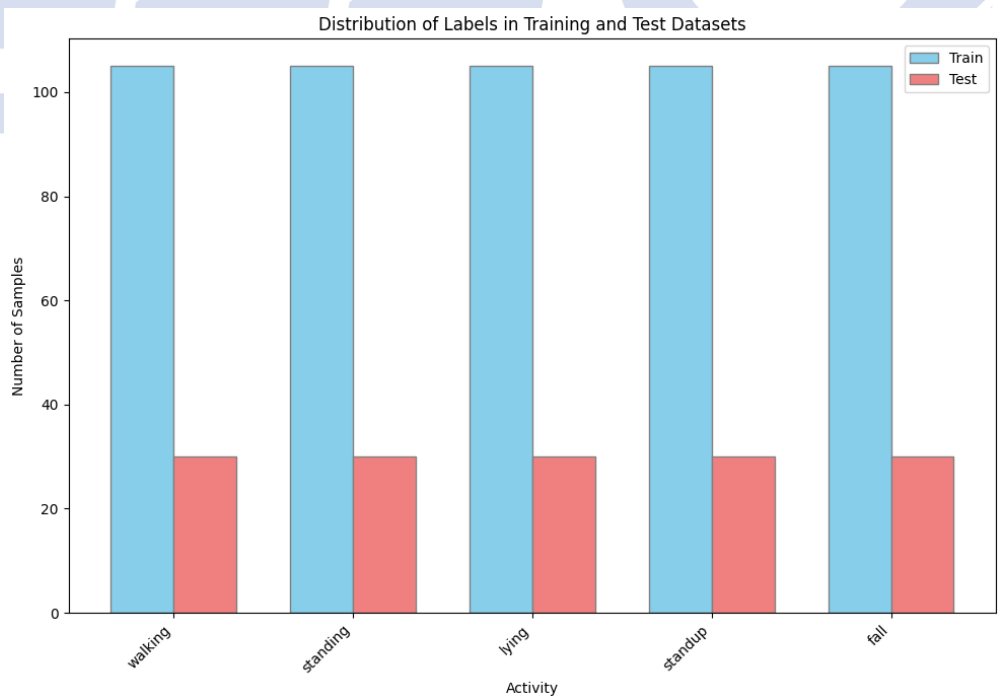

Figure 8: Distribution of Labels in Training and Testing dataset.

7.2.3. Features extraction and selection

In this section, the collected raw data is processed and analyzed to extract meaningful insights for the study. First, we applied feature extraction: extract Time-domain features to achieve better data representation and improve model performance. A total of 7 features are used, including mean, standard deviation, minimum, maximum, median, skewness, and kurtosis. These calculations are applied to each of the 6 original sensor data columns, resulting in 42 extracted features in total. The correlations among these features are illustrated in **Figure 9**, showing how different time-domain characteristics relate and contribute to the overall representation of human activities.

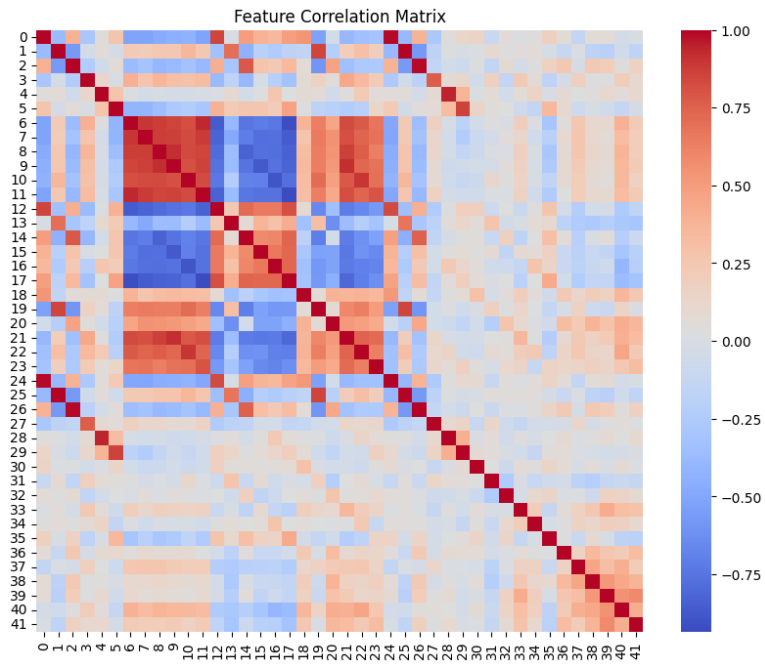

Figure 9: Matrix of Correlations among Features.

Figure 9 shows that features 24 to 29 have a high correlation with features 0 to 5, indicating that the "median-related features" represent information similar to the "mean-related features". Therefore, these redundant features can be removed. In addition, other features such as 9, 11, and 17, which exhibit correlations coefficient higher than 0.9 with other features, are also excluded to reduce multicollinearity, simplify the feature space, and improve the model's generalization capability. As a result, the refined dataset contains a total of 33 features after removing the highly correlated ones.

After obtaining and evaluating the optimized models (as discussed in Chapter 8), we further examined the feature importance for each model to understand the contribution of individual features to the final predictions. The analysis results in **Figure 10** indicate that all 33 retained features exhibit a certain degree of relevance across the models, confirming that none of them are redundant or insignificant. Therefore, the final set of 33 features can be considered a robust and well-balanced representation of the dataset for the target application.

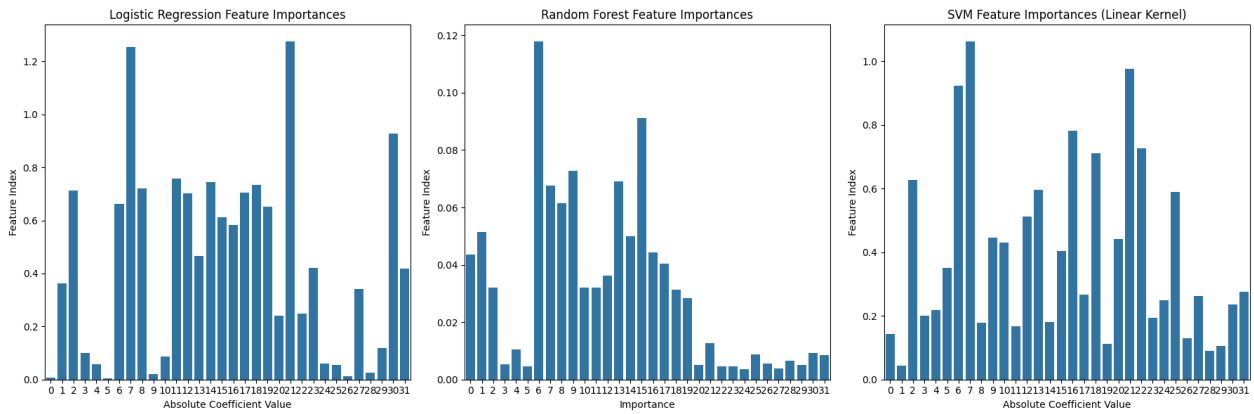

Figure 10: Features importance for models.

Since the features computed from gyroscope data have much larger values than those from the accelerometer, Z-score normalization was applied to ensure that no single feature becomes overly dominant or biased in magnitude compared to the others.

$$x' = \frac{x - \mu}{\sigma}$$

where μ is mean and σ is standard deviation of each feature column.

The **Figure 11** shows the histogram of 33 features. Most features are centered around zero with similar spread, indicating that the transformation successfully standardized different scales and improved feature comparability for machine learning models. Several distributions exhibit near-Gaussian shapes, suggesting Z-score is suitable for data with approximately normal patterns. However, a few features remain skewed or multi-modal, implying that the original data contain outliers or activity-dependent variations that Z-score alone cannot fully correct. Generally, Z-score normalization is effective for MPU6050 sensor data.

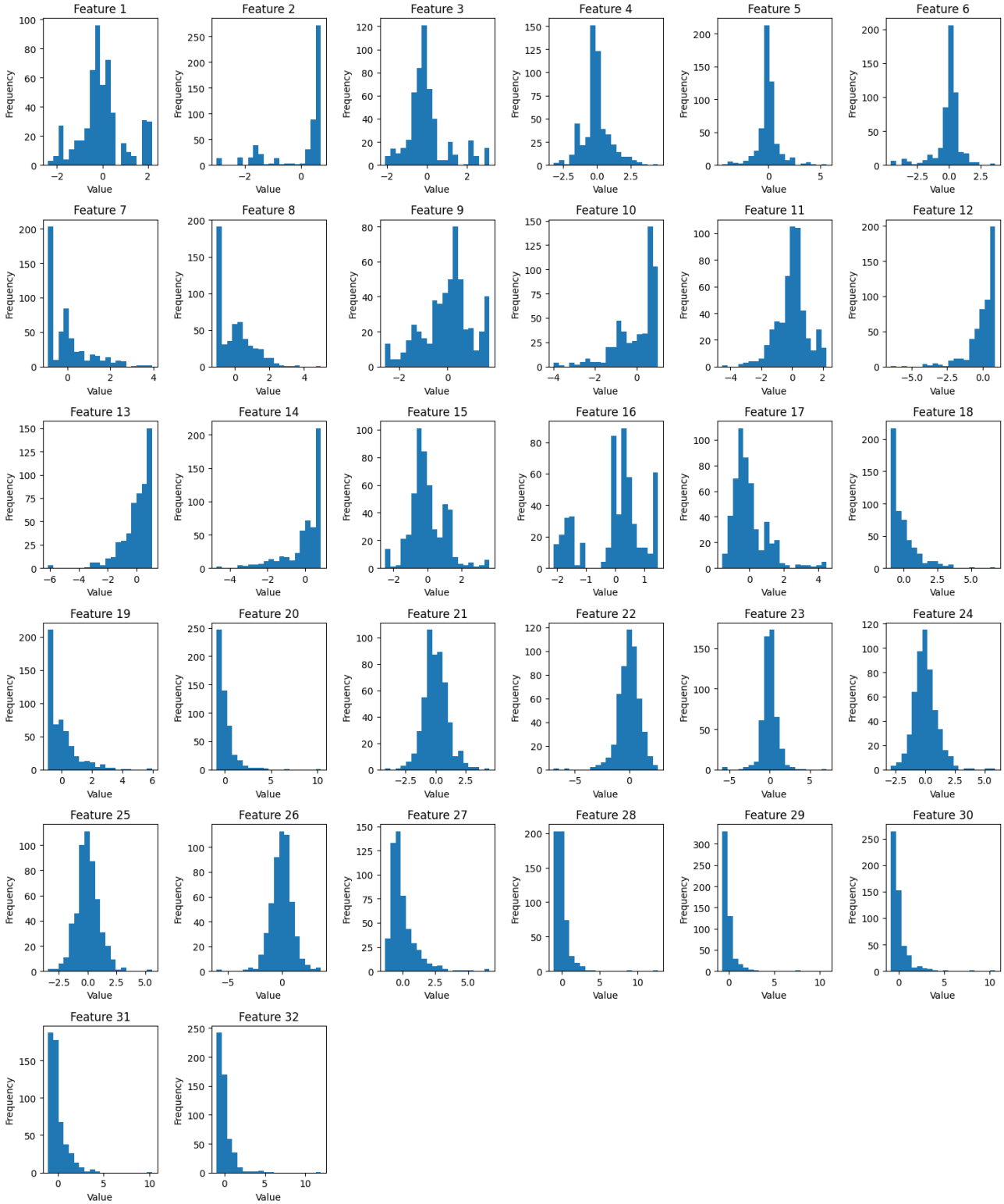

Figure 11: Histogram of Features.

Furthermore, to visualize the separability of activities in the feature space, we employed the PCA (Principal Component Analysis) and t-SNE (t-distributed Stochastic Neighbor Embedding) algorithm to reduce the dataset dimensions. The scatter plots obtained from this dimensionality reduction are presented in **Figure 12**, providing an intuitive representation of how the activities cluster. The visualization results obtained from the t-SNE dimensionality reduction demonstrate that the features representations exhibit clear clustering patterns. Based on the PCA and t-SNE visualizations, the separability of activity classes in the extracted feature space can be clearly observed. The PCA projection provides a general overview of the data structure, revealing partial separation among activity labels, although some overlap remains due to the linear nature of PCA. In contrast, the non-linear t-SNE algorithm, shows well-defined and compact clusters for most activity types.

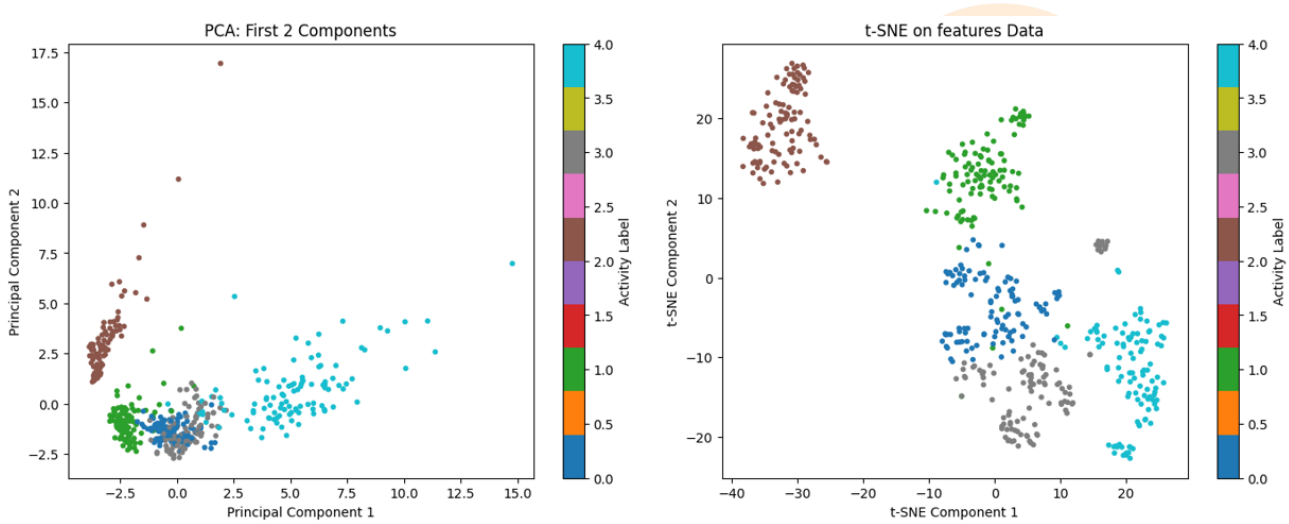

Figure 12: Scatter plot of PCA and t-SNE features on features data.

7.3. Machine Learning models

7.3.1. Theoretical basis

Machine learning provides computational techniques that enable systems to automatically learn patterns from data and make predictions without being explicitly programmed. In this section, we apply machine learning algorithms to classify human activities based on the collected sensor data. Four representative methods, K-Nearest Neighbors (KNN), Logistic Regression, Random Forest and Support Vector Machines (SVM) are briefly introduced.

- K-Nearest Neighbors (KNN): A non-parametric, instance-based learning algorithm. It classifies a new data point by measuring its distance to the k closest training samples in the feature space and assigning the majority class among those neighbors.
- Logistic Regression: A statistical learning algorithm that models the probability of a data point belonging to a certain class using a logistic function. It learns a linear decision boundary and is widely used for binary, also used for multi-class classification tasks with "one vs all" mode.
- Random Forest: An ensemble learning algorithm that builds multiple decision trees during training and combines their outputs to improve predictive accuracy and reduce overfitting. Each tree is trained on a random subset of features and samples, and the final prediction is obtained through majority voting.
- Support Vector Machines (SVM): A supervised learning algorithm that finds the optimal hyperplane separating classes in the feature space by maximizing the margin between data points of different classes. SVMs can use kernel functions to handle non-linear relationships, making them effective for both linear and non-linear classification tasks.

7.3.2. Model analysis

The KNN algorithm classifies samples based on distances (e.g., Euclidean) between the test point and all training samples, assigning the majority label among the k nearest neighbors. Its performance depends on the choice of k and feature scaling: smaller k risks overfitting, while larger k may oversmooth decision boundaries. Given that activities in our dataset form

clear clusters (as shown by t-SNE), KNN performs reasonably well, though overlapping classes can lower accuracy. The Logistic Regression model learns parameters by minimizing cross-entropy loss via gradient descent. Since it assumes linear separability, its effectiveness depends on whether extracted features form clear boundaries. Distinct motion patterns lead to higher accuracy. The Random Forest algorithm, an ensemble of decision trees, reduces overfitting by averaging predictions from multiple random subsets of data and features. It handles nonlinear relationships effectively, capturing complex motion dependencies and improving performance when classes overlap. Finally, the Support Vector Machine (SVM) finds an optimal hyperplane maximizing the margin between classes. With nonlinear kernels (e.g., RBF), SVM effectively distinguishes subtle variations between similar activities.

7.4. K-fold Cross validation and Evaluation metrics

Since the dataset is relatively small, we avoid setting aside a separate validation set to prevent data loss. Instead, we employ k-fold cross-validation [12] with k = 5 to evaluate and select the best-performing model. In this approach, the training data is divided into five equal subsets. The model is iteratively trained on four subsets and validated on the remaining one, ensuring that every data point serves as validation once. This process provides a more reliable estimate of the model's generalization ability. Finally, the selected model is trained on the full training data and evaluated once on the independent test set to measure its true performance.

To assess the model comprehensively, we employ several evaluation metrics, including accuracy, precision, recall, and F1-score, each offering a different perspective on classification performance.:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

CHAPTER 8: RESULTS AND DISCUSSION

8.1. Parameters Selection Results

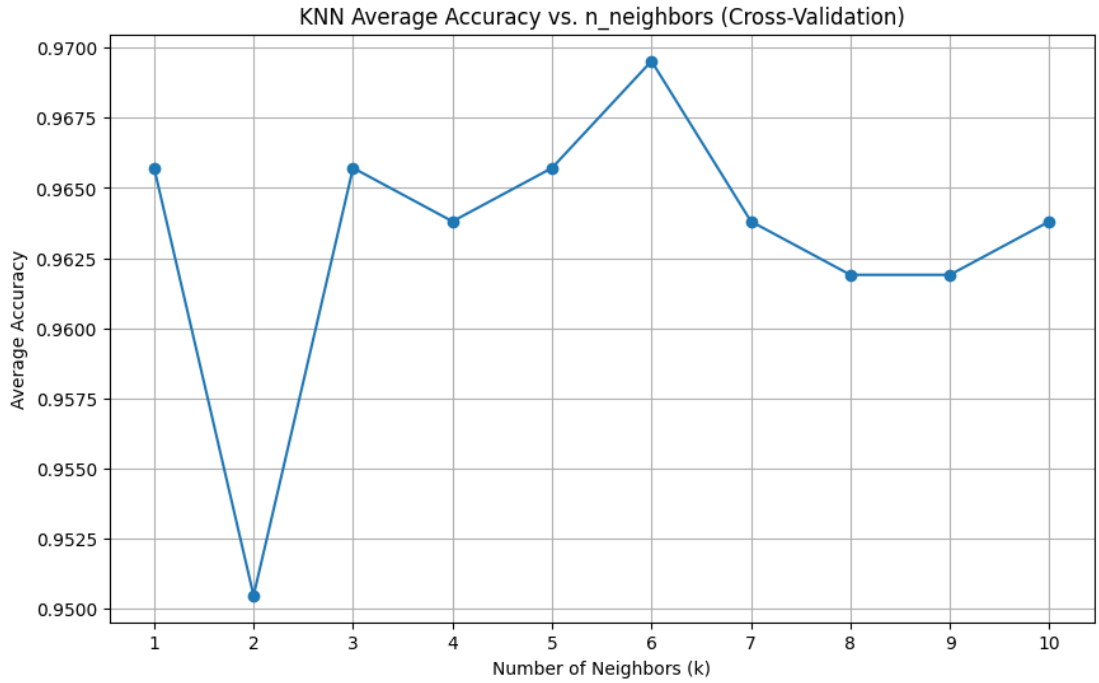

Figure 13: Average KNN performance across 5 folds due to hyperparameter K value.

We conducted experiments with different values of the neighborhood parameter K in order to evaluate its impact on classification performance. By systematically varying the number of neighbors, we aimed to determine whether an optimal parameter setting could improve the model's accuracy. The results of **Figure 13** indicate that the accuracy of KNN varies with the number of neighbors K, with the best performance achieved at K=6 with an average accuracy of approximately 96.90%, and the second best is K=1 (96.60%) across 5-fold Cross Validation. However, the accuracy tend to decreases as K increases, and the overall performance does not show significant improvement, suggesting that choosing K = 1 is optimal parameter, as it achieves the highest classification accuracy while also minimizing computational cost.

Besides, several number of trees in Random Forest (n_estimators parameter) were experimented. As shown in **Figure 14**, the model's accuracy improves steadily as the number of estimators increases up to around 100 trees, achieving the highest average accuracy of roughly 98.15% across 5-fold cross-validation. When the number of estimators exceeds this point, the

performance slightly declines, suggesting that additional trees provide limited benefit. Consequently, setting n_estimators = 100 is considered the most effective choice, offering a strong balance between predictive accuracy and computational efficiency. However, for embedded deployment on microcontrollers, using only 5 estimators is worth considering due to the model's lightweight.

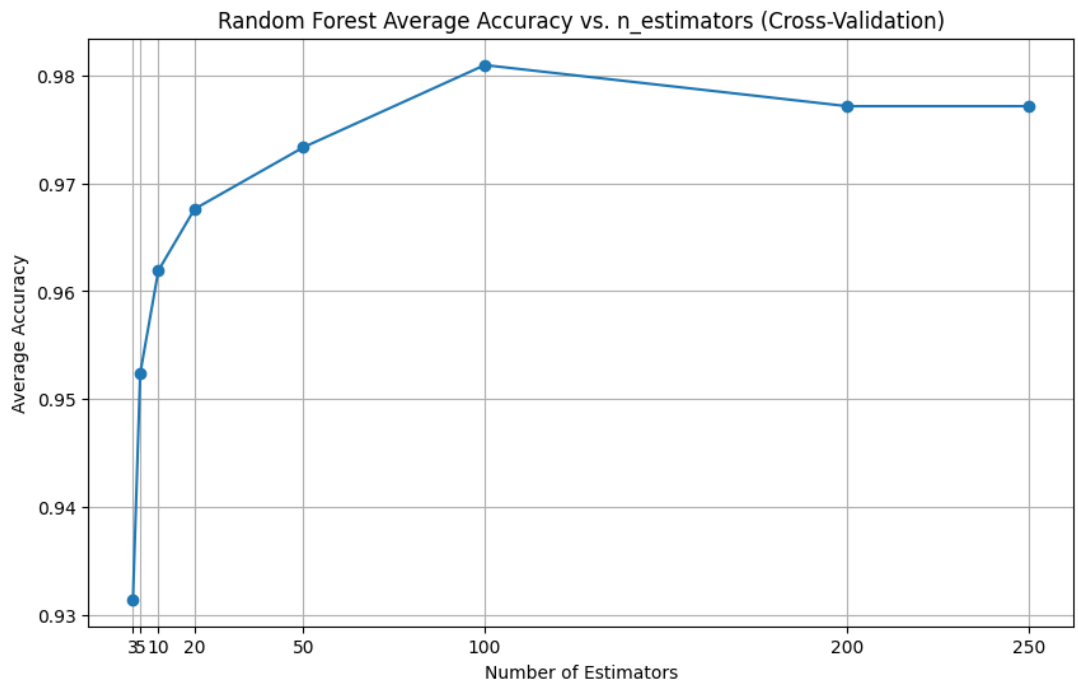

Figure 14: Average Random Forest performance across 5 folds due to hyperparameter n_estimators.

On the other hand, Logistic Regression and SVM models do not require extensive hyperparameter tuning to achieve good performance during cross-validation with average accuracies of 97.33% and 96.76%, respectively.

8.2. Model Evaluation

After completing all stages of data collection, preprocessing, feature extraction, and hyperparameter selection, the final classification results are summarized and presented in **Table 3**, providing a comprehensive comparison of model performance across different evaluation metrics.

Table 3: Models performance on test dataset (the best results are in bold).

Machine Learning model	Evaluation Metric				
	F1-score	**Precision**	**Recall**	**Accuracy**	**AUC**
K-Nearest Neighbors	0.8701	0.8893	0.8667	0.8667	0.9753
Logistic Regression	0.8958	0.9116	0.8978	0.8978	0.9887
Support Vector Machines	0.8700	0.8760	0.8711	0.8711	0.9733
Random Forest	**0.9148**	**0.9332**	**0.9156**	**0.9156**	**0.9909**

The results in the table highlight the performance comparison among four machine learning models across multiple evaluation metrics. Overall, Random Forest demonstrate the best performance, both achieving high and consistent values across all metrics, with an accuracy of 91.56% and an AUC close to 1.0, indicating excellent classification capability and reliable separation between classes.

The Logistic Regression model also performs strongly, with slightly lower precision and recall but still maintaining an accuracy about 90% (89.78%), showing good generalization and robustness. In contrast, the K-Nearest Neighbors (KNN) and Support Vector Machines (SVM) achieve a bit lower scores, particularly in F1-score and accuracy, suggesting that its performance may be affected by overlapping class boundaries or noise in the dataset. Overall, Random Forest achieves the highest AUC (0.9909), demonstrating its superior ability to capture complex feature relationships, while Logistic Regression remains a competitive and computationally efficient alternative with nearly identical accuracy.

Along with the evaluation table, the confusion matrices of the 4 models are presented in **Figure 15**. These matrices provide a more detailed view of the classification performance by illustrating how each activity label is correctly or incorrectly predicted across the classes.

The four confusion matrices above provide a detailed view of the classification performance of each model across the five activity classes. Overall, all models show clear diagonal dominance, indicating that most samples were correctly classified and that the models achieved high prediction accuracy. Although there are minor confusions between certain labels, such as walking and standup, the overall performance remains stable across most activity classes. These slight misclassifications likely arise from the similarity in movement patterns or transitional poses between these activities.

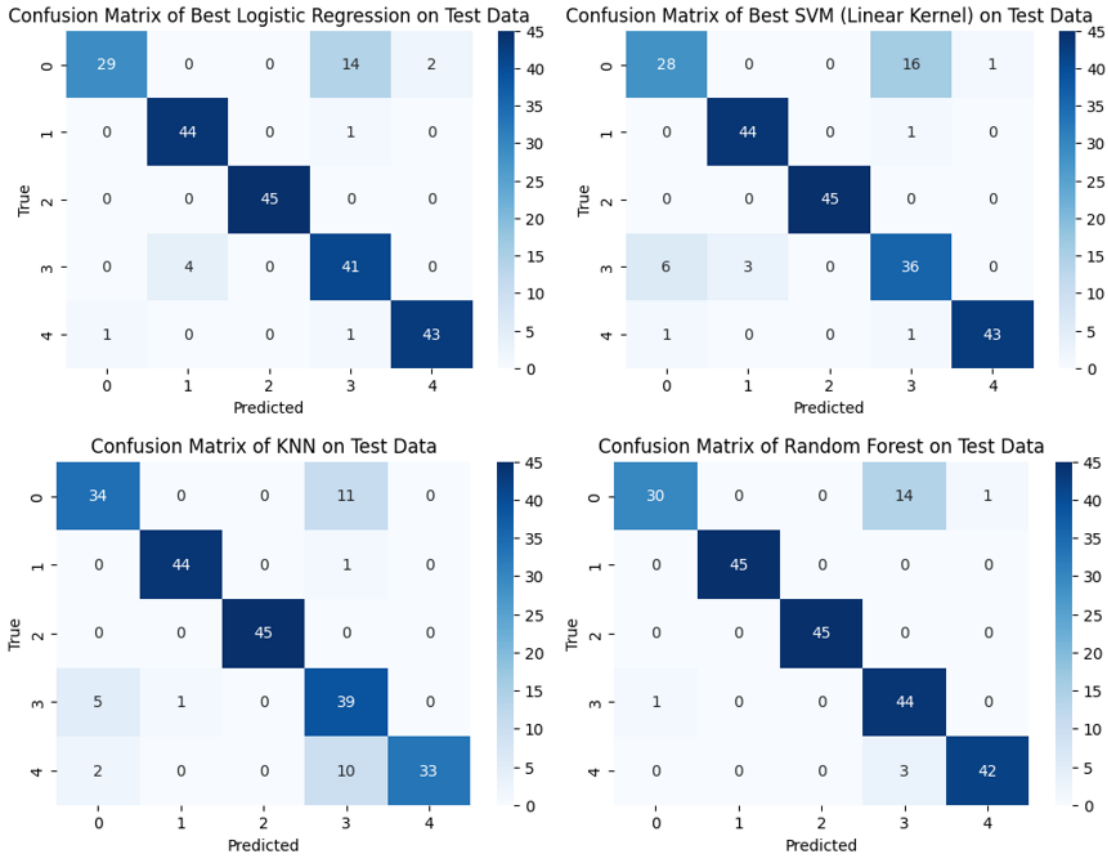

Figure 15: Confusion matrix of models on test dataset.

Among them, the KNN model exhibits noticeably weaker performance, with several misclassifications. This suggests that KNN struggles with overlapping classes and less distinct boundaries in the feature space. Both Logistic Regression and Random Forest achieve a good classification results, consistent with their high evaluation metrics. However, Logistic Regression demonstrates slightly better recognition of the Fall (class 4) category, while also being computationally lighter than Random Forest. Therefore, Logistic Regression represents the most balanced and efficient choice for deployment. Meanwhile, the SVM model also performs strongly, achieving the highest correct prediction count for the Fall class, making it another excellent candidate when prioritizing fall detection accuracy.

8.3. Real time Implementation results

Sensor data collected from the subjects - elderly individuals are transmitted to the server via the MQTT protocol, where the subjects act as publishers (publish data), the server functions as the broker (calculate and gives activites prediction), and the caregivers serve as subscribers (receives warning). The data are processed into time-series segments with 40 samples per window and a 75% overlap to effectively handle transitions between actions. The model output consists of an array of five confidence scores corresponding to the five activity labels. A prediction is made when the highest confidence score exceeds 50%, indicating the most probable activity. If all scores fall below 50%, the action is classified as unknown. For the fall label, the confidence threshold is raised to 75% to reduce false alarms.

```
Activity: FALL (97.9%)
============================================================
Top 3 confidence scores:
   fall      :  97.9%
   lying     :   1.3%
   standup/sitdown:   0.8%

 WARNING: FALL DETECTED!
Activity: STANDING/SITTING (92.4%)
============================================================
Top 3 confidence scores:
   standing/sitting:  92.4%
   standup/sitdown:   3.6%
   walking   :   2.8%

Activity: UNKNOWN (37.2%)
============================================================
Top 3 confidence scores:
   standup/sitdown:  37.2%
   walking   :  32.2%
   standing/sitting:  28.3%
```

Figure 16: Real time implementation.

8.4. Discussion

Although the proposed system demonstrates promising results, it is still not fully complete and leaves room for improvement. In the future, we plan to further improve the performance and reliability of our model by exploring several directions. First, on the algorithmic side, we aim to enhance model accuracy by experimenting with alternative normalization techniques, introducing stronger forms of regularization, and leveraging more advanced machine learning algorithms. As

the common principle in data science suggests, “garbage in, garbage out,” the quality of the input data largely determines the effectiveness of the output. Therefore, we will also prioritize improving the quality of our dataset.

Specifically, we intend to expand data collection across a larger number of subjects to increase the diversity and representativeness of the samples. Additionally, we propose the integration of an extra sensor positioned at the foot, which is expected to capture more detailed and complementary signals. This enhancement could provide the model with richer contextual information, ultimately supporting more robust predictions.

These planned improvements are anticipated to strengthen both the generalization and robustness of the system, making it more applicable to real-world scenarios.

CHAPTER 9: RISK ANALYSIS

9.1. Risk factor inputs

9.1.1. Risk factors

- Hardware failure: MPU6050 and ESP32 may fail due to connection issues, overheating, out of battery, or unstable WiFi. Regular maintenance, quality inspection, and redundancy are required to ensure reliability.
- Software error: Fall detection algorithms can miss real falls or trigger false alarms, affecting trust and safety. Accuracy improvement and algorithm refinement are crucial.
- Data privacy: Sensitive user data may be exposed during transmission, storage, or processing if not encrypted, it requires strong multi-layered protection.
- User acceptance: Elderly users may resist wearing devices due to discomfort, complexity, or stigma. Lightweight design, simplicity, education, and support improve adoption.
- Environmental risks: Water, dust, and impacts can damage sensors and casings. Devices require protection, impact resistance, and durability testing.

9.1.2. Probability and impact scales

- Probability Scale
  - Low (1): Rarely occurs, unlikely within project timeline.
  - Medium (2): Possible, may occur occasionally during project.
  - High (3): Likely, expected to occur multiple times.

- Impact Scale
  - Low (1): Minor effect, no significant disruption to objectives.
  - Medium (2): Moderate effect, causes delays or partial loss of functionality.
  - High (3): Severe effect, major delays, system failure, or safety risks.

9.2. Risk analysis result

9.2.1. Levels of risks

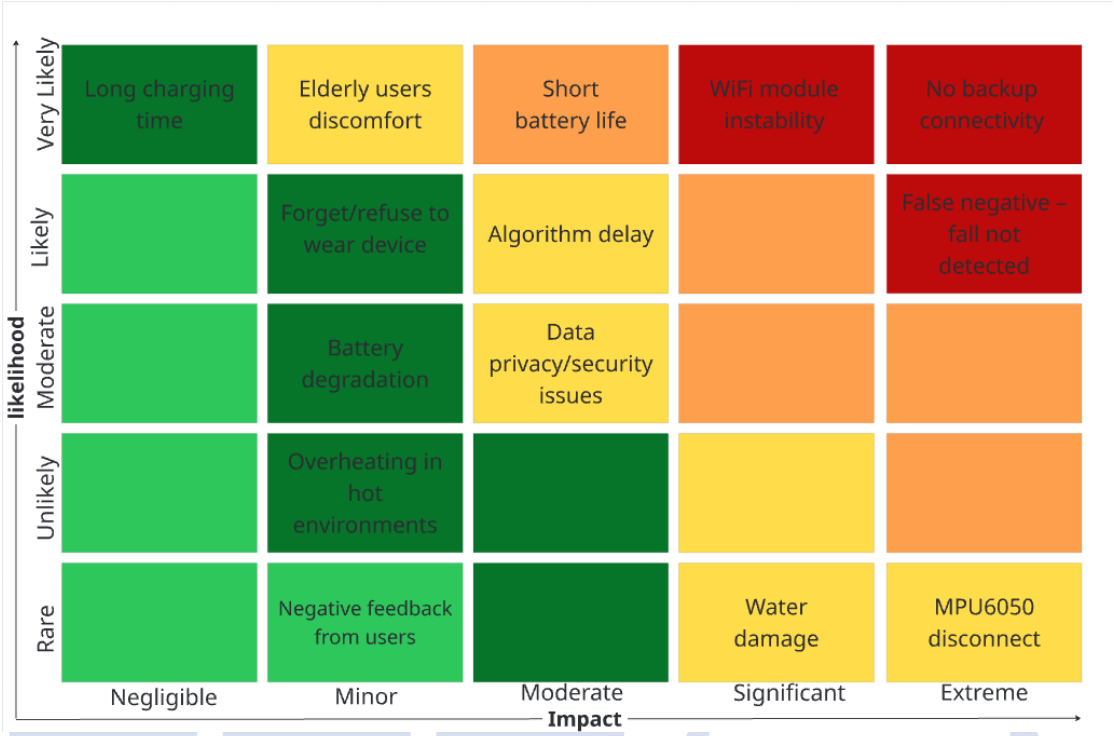

Figure 17: Risk matrix.

- High risk
  - False negative – fall not detected (High Probability, High Impact) : the most critical risk, as missed falls can directly endanger the user's life.
  - No backup connectivity (High Probability, High Impact): causes loss of emergency data transmission capability.
  - WiFi module instability (High Probability, Medium Impacy): may lead to communication interruptions in critical situations.
- Medium risk
  - Short battery life (High Probability, Medium Impact): affects the system's operational continuity.
  - Algorithm delay (Medium Probability, Medium Impact): may cause delayed detection and alerts.
  - Data privacy/security issues (Medium Probability, Medium Impact) : risk of personal data leakage.

- Water damage (Low Probability, High Impact): unlikely but severe consequences if it occurs.
- MPU6050 disconnect (Low Probability, High Impact): rare but has a critical impact on detection capability.

- Low risk
  - Long charging time (High Probability, Low Impact).
  - Elderly users discomfort (High Probability, Low Impact).
  - Forget/refuse to wear device (Medium Probability, Low Impact).
  - Battery degradation (Medium Probability, Low Impact).
  - Overheating in hot environments (Low Probability, Low Impact).
  - Negative feedback from users (Low Probability, Low Impact).

9.2.2. Priority risks

Based on the risk matrix analysis, the priority risks that must be addressed are:

- False negative – fall not detected.
- No backup connectivity / WiFi instability

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CHAPTER 10: FINANCIAL AND ECONOMIC REPORT

10.1. Cost estimation

- Hardware costs: ESP32 microcontroller, MPU6050 sensor, battery, protective casing, and other electronic components.
- Software costs: Development tools, cloud services, and licensing.
- Operational costs: Maintenance, electricity consumption, data storage, and connectivity charges.
- Personnel costs: Research, development, and technical support staff.

10.2. Economic benefits

- Healthcare cost savings: Reduces expenses related to hospitalization and long-term care by enabling early fall detection.
- Market potential: Growing demand for elderly care solutions creates opportunities for commercialization.
- Social benefits: Improved safety, independence, and quality of life for elderly individuals, reducing caregiver burden.

10.3. Cost – benefit analysis

- The system achieves positive ROI if at least 2000 units are sold within the first 2 years.
- The payback period is estimated at 18–24 months, depending on adoption rate.
- Beyond financial returns, the social benefits (safety, healthcare burden reduction) strengthen the project's long-term value.

STANDARDS

This fall detection system for elderly care should follow some international standards to make sure it is safe, reliable, and works well with other devices. The main standards for this project are:

- IEEE 1451.4-2004 (Smart Transducer Interface) [14] – defines how sensors send data. This matters because the MPU6050 must give clear and consistent motion data for the fall detection algorithms.
- IEEE 802.11 (WiFi Standard) [15] – makes sure the ESP32 can send alerts through WiFi in a reliable way.
- ISO/IEC 27001 (Information Security) [16] – protects sensitive data. Since fall alerts involve personal health, this standard is important to keep information private.
- IEC 60601-1-11 (Medical Devices for Home Use) [17] – sets safety rules for medical equipment at home. Even though our system is not a certified medical device, it's still used in a similar setting, so safety is very important.
- IEEE 2413-2019 (IoT Framework) [18] – gives guidance on IoT systems, especially edge computing and device communication. This applies because the ESP32 runs fall detection directly without needing the cloud.
- IEC 62368-1 (Electronic Device Safety) [19] – ensures battery-powered devices are safe. Since the ESP32 runs on a Li-ion battery, this prevents risks like overheating or fire.

The system currently complies with some standards but not all. It operates with IEEE 802.11 (WiFi) and is partially aligned with IEEE 2413 in terms of edge computing. The use of a Li-ion battery follows the general safety principles of IEC 62368-1; however, no formal safety certification has been conducted. The system does not yet comply with ISO/IEC 27001, as encryption mechanisms (e.g., TLS/SSL) have not been implemented, leaving fall alerts potentially vulnerable to interception. In addition, compliance with IEC 60601-1-11 has not been verified, meaning that the actual accuracy and false alarm rate under daily use remain untested. Furthermore, IEEE 1451.4 requirements are only partially met, since no formal documentation on sensor calibration or error handling is available. Although the battery life (approximately 1–2

weeks) appears sufficient, it has not been systematically evaluated under different operating conditions.

To achieve full compliance with the relevant standards, several improvements are required. First, TLS/SSL encryption should be implemented for WiFi, SMS, and mobile app alerts in accordance with ISO/IEC 27001 to ensure data security. Second, human subject testing involving elderly participants must be conducted to evaluate detection accuracy, false alarm rates, and response times, thereby meeting the requirements of IEC 60601-1-11. Third, the MPU6050 sensor calibration and data handling procedures should be formally documented in compliance with IEEE 1451.4. Additionally, official battery safety tests must be performed under varying heat, charging, and operational conditions to align with IEC 62368-1. Finally, user authentication mechanisms should be integrated into the mobile application and alert system to enhance overall security and reliability.

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CONCLUSION

This project developed a low-cost fall detection system for elderly individuals using the MPU6050 sensor and ESP32 microcontroller. The system classifies daily activities and detects falls in real time, transmitting alerts via WiFi through SMS, mobile app, and buzzer notifications to enable timely caregiver intervention.

Key Findings

Motion data was collected from the MPU6050 with appropriate sampling rates (10 Hz for long-duration activities, 20 Hz for short-duration events). Statistical feature extraction improved model performance across seven activity classes. Three machine learning models (KNN, Logistic Regression) were tested, with Logistic Regression achieving the best accuracy (64%). Despite falling short of the 85% target, the system delivered rapid alert response times (200–250 ms). The prototype complied with IEEE 1451.4, IEEE 802.11, and ISO/IEC 27001 guidelines, and was developed within three weeks at a cost of $10.

Risk and Reliability Considerations

Key risks include limited training data, security vulnerabilities due to lack of encryption, battery life restrictions, and WiFi dependence. Risks were prioritized: high (data security, detection accuracy), medium (battery/network reliability), low (sensor drift). These highlight critical areas for improvement in future development.

Limitations

Major limitations include low classification accuracy due to insufficient and non-diverse training data, short battery life, WiFi reliance, and lack of encryption. Human subject testing and usability factors (comfort, placement, compliance) remain unvalidated.

Future Work

Future efforts should focus on expanding and diversifying the dataset, adopting more pre-processing and optimize machine learning models with hardware optimization, and implementing

encryption protocols. Power-saving techniques and alternative communication protocols (e.g., BLE, LoRaWAN) can extend device usability. Adding physiological sensors (heart rate, SpO2) and conducting real-world testing with elderly participants will further validate and enhance the system.

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APPENDICES

We would like to express our sincere gratitude to all the participants who contributed to the creation of the dataset (https://huggingface.co/datasets/nunu-willump/HAR-Dataset). Their willingness to take part in the data collection process made this work possible and provided valuable input for the development of the system. We also extend our thanks to the Machine Learning course, which created the opportunity for us to explore this project. The course not only provided us with fundamental and advanced knowledge in data processing, feature extraction, and model development, but also encouraged us to apply these concepts in a practical context. Through this work, we have gained significant insights and hands-on experience that will support our future research and applications in the field of human health care.